THE PROJECTION FORMULA FOR A PROPER MORPHISM OF SMOOTH SCHEMES OVER A FIELD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FFC (sources/stacks-chow/pinned-original-source.tex line 14482).

Allowed standard prerequisites (editorial).

- Existence of the Gysin map for a local complete intersection morphism. Same chapter, Definition labelled definition-lci-gysin, cited at proof lines 14490-14494. Provides f^! for the proper morphism f.
- Identification of Chow classes of smooth schemes with caps of bivariant classes. Same chapter, Lemma labelled lemma-identify-chow-for-smooth, cited at proof lines 14501-14507 and 14517-14519. Expresses alpha and beta as caps.
- The value of f^! on the fundamental class and its commutation with bivariant classes. Same chapter, Lemma labelled lemma-lci-gysin-easy, cited at proof lines 14499-14501, and Lemma labelled lemma-lci-gysin-commutes, cited at proof lines 14509-14513. Supplies f^![Y]_e = [X]_d and lets c' be moved past f^!.
- Bivariant classes commute with proper pushforward. Same chapter, the definition of a bivariant class, used at proof lines 14513-14517. Allows c' to be moved past f_*.

Source reference. Original definition 0FF3: sources/stacks-chow/pinned-original-source.tex, source lines 13530–13530. Same-chapter pointer into the author environment definition at source line 13530; invoked at proof line 14491 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FF7: sources/stacks-chow/pinned-original-source.tex, source lines 13643–13643. Same-chapter pointer into the author environment lemma at source line 13643; invoked at proof line 14496 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

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Source reference. Original lemma 0FC2: sources/stacks-chow/pinned-original-source.tex, source lines 14403–14403. Same-chapter pointer into the author environment lemma at source line 14403; invoked at proof line 14498 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FF6: sources/stacks-chow/pinned-original-source.tex, source lines 13613–13613. Same-chapter pointer into the author environment lemma at source line 13613; invoked at proof line 14515 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let k be a field. Let $f : X \to Y$ be a proper morphism of schemes smooth over k. Then the Gysin map exists for f and $f_*(\alpha \cdot f^!\beta) = f_*\alpha \cdot \beta$.*

Proof. Observe that $X \to X \times_k Y$ is an immersion of X into a scheme smooth over Y. Hence the Gysin map exists for f (Definition 0FF3). To prove the formula we may decompose X and Y into their connected components, hence we may assume X is smooth over k and equidimensional of dimension d and Y is smooth over k and equidimensional of dimension e. Observe that $f^![Y]_e = [X]_d$ (see for example Lemma 0FF7). Write $\alpha = c \cap [X]_d$ and $\beta = c' \cap [Y]_e$, see Lemma 0FC2. We have

$$\begin{aligned} f_*(\alpha \cdot f^!\beta) &= f_*(c \cap f^!(c' \cap [Y]_e)) \\ &= f_*(c \cap c' \cap f^![Y]_e) \\ &= f_*(c \cap c' \cap [X]_d) \\ &= f_*(c' \cap c \cap [X]_d) \\ &= c' \cap f_*(c \cap [X]_d) \\ &= \beta \cdot f_*(\alpha) \end{aligned}$$

The first equality by the result of Lemma 0FC2 for X. By Lemma 0FF6 we know that $f^!$ commutes with c'. The commutativity of the intersection product justifies switching the order of capping $[X]_d$ with c and c' (via the lemma). Commuting c' with f_* is allowed as c' is a bivariant class. The final equality is again the lemma. □